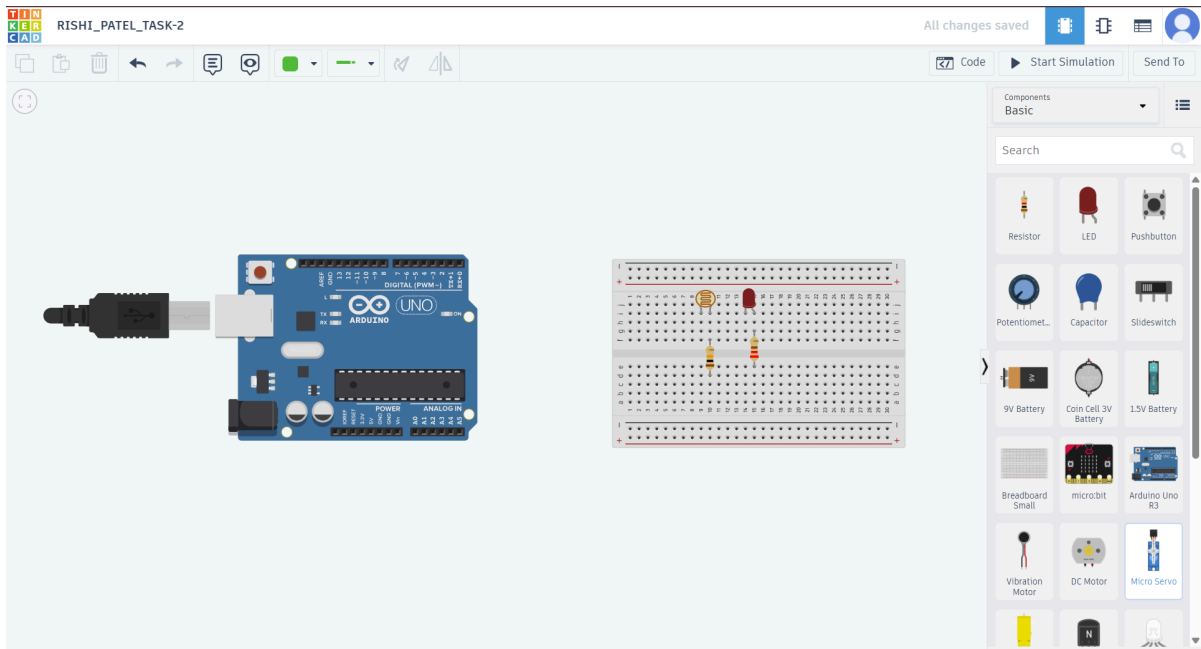


TASK_2

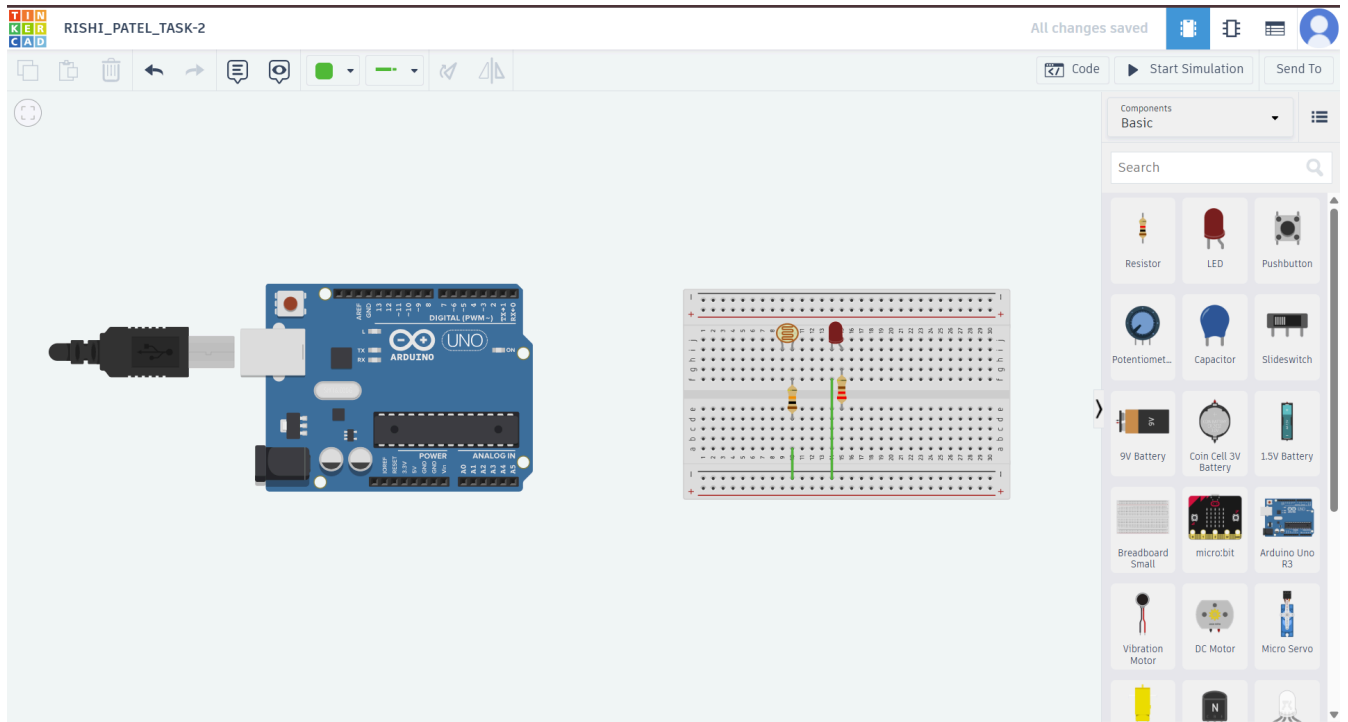
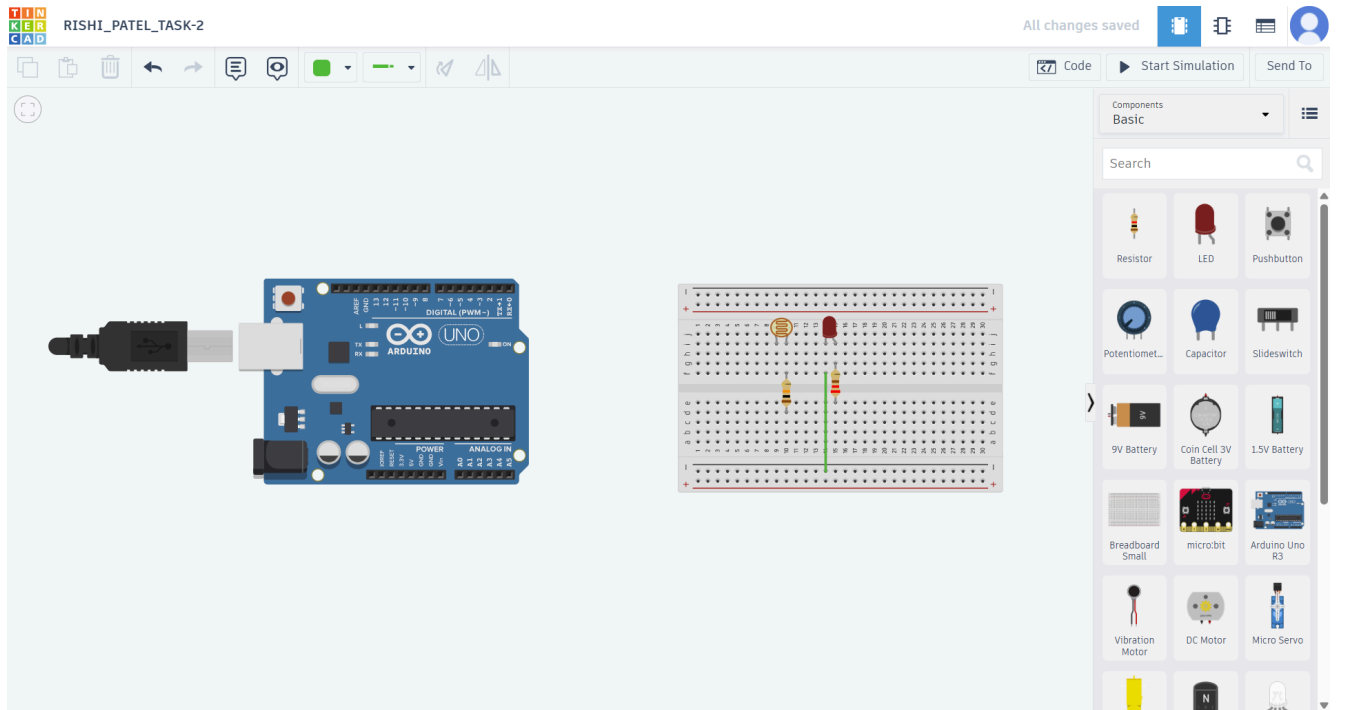
In This Task We Will See How To Perform LED On/Off Using Light Sensor.

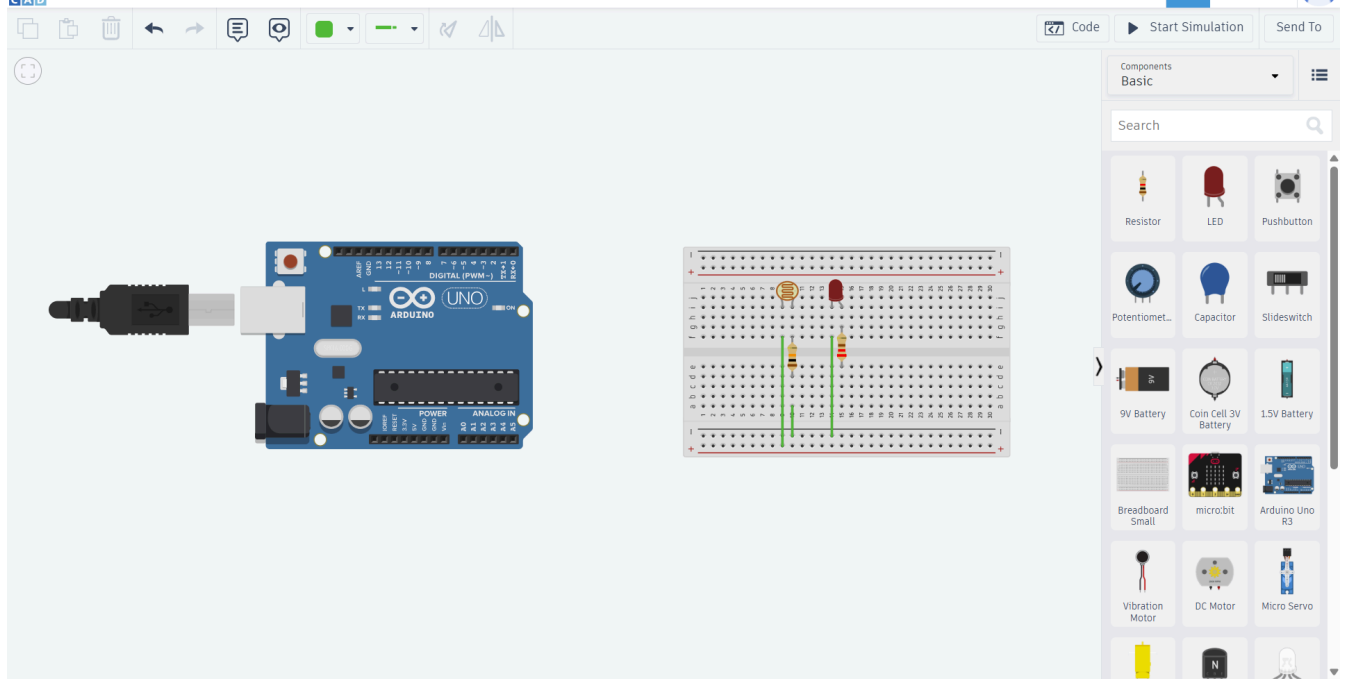
This Will Be Shown On The Basis Of The Screenshots And The Code Which Is Important For The Task

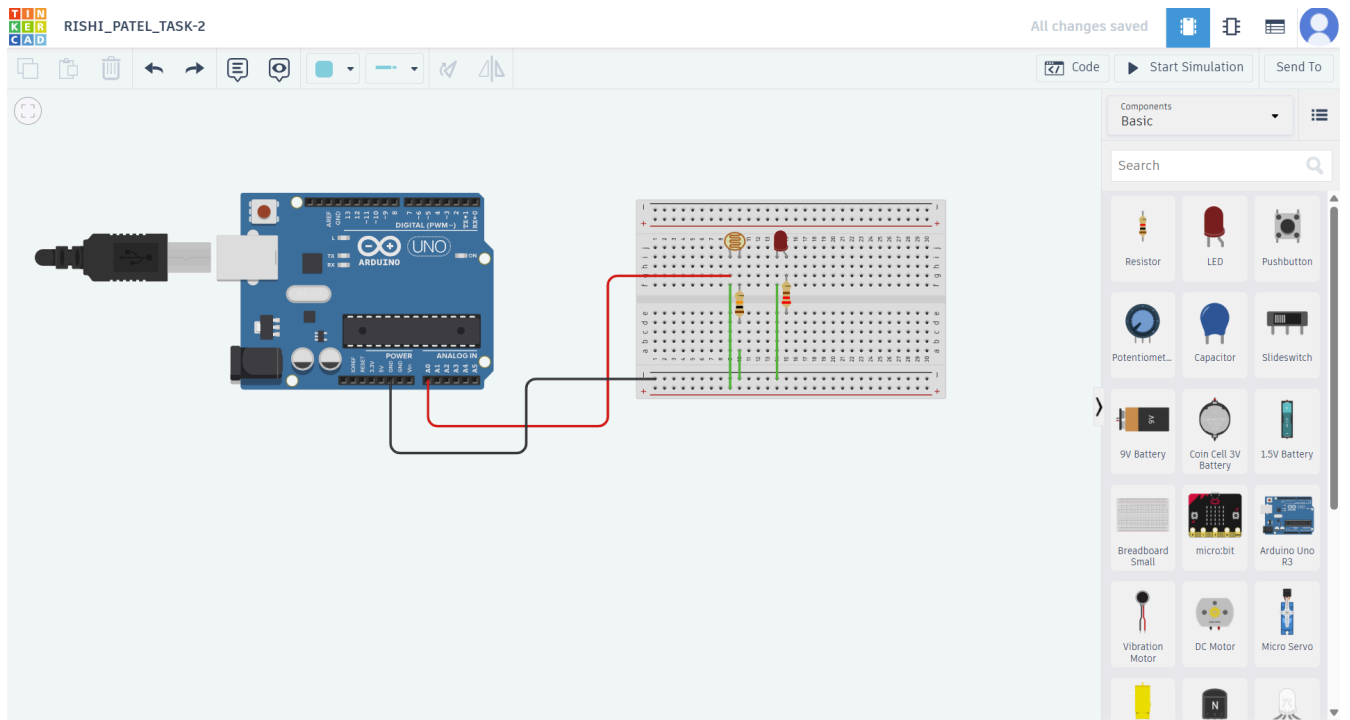
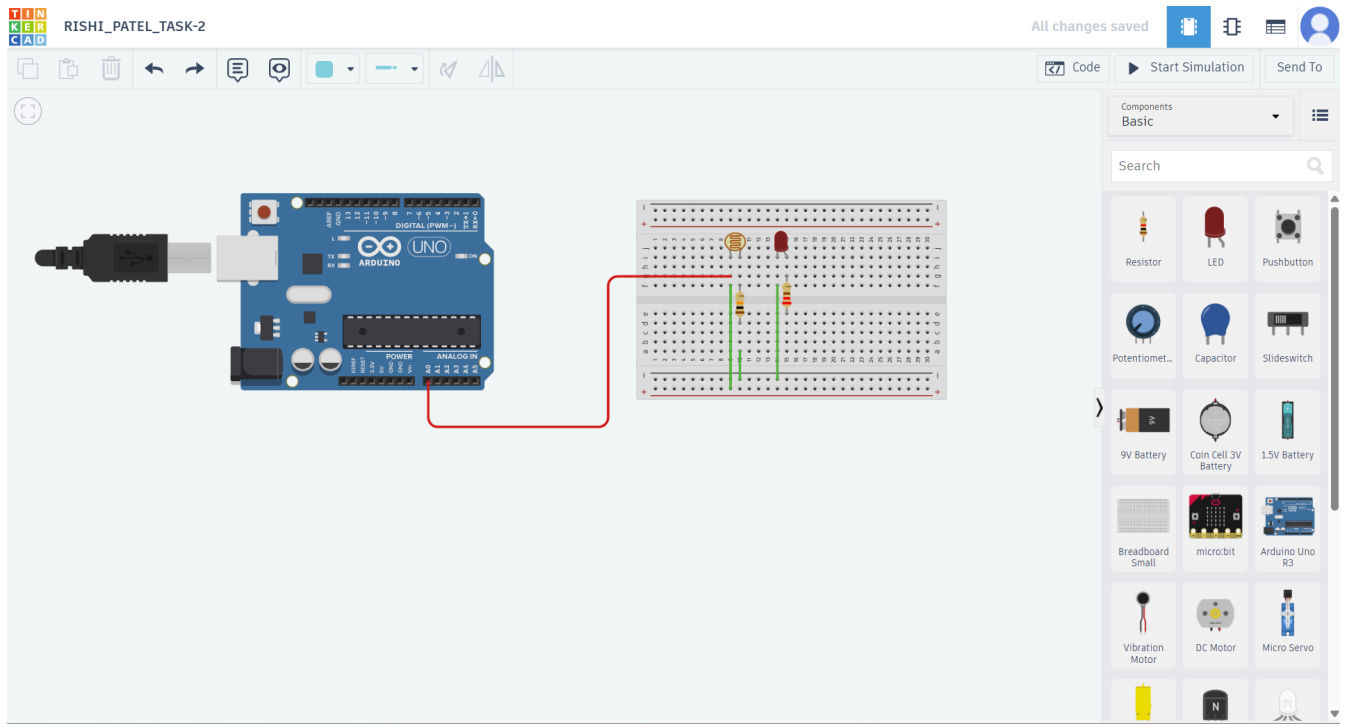


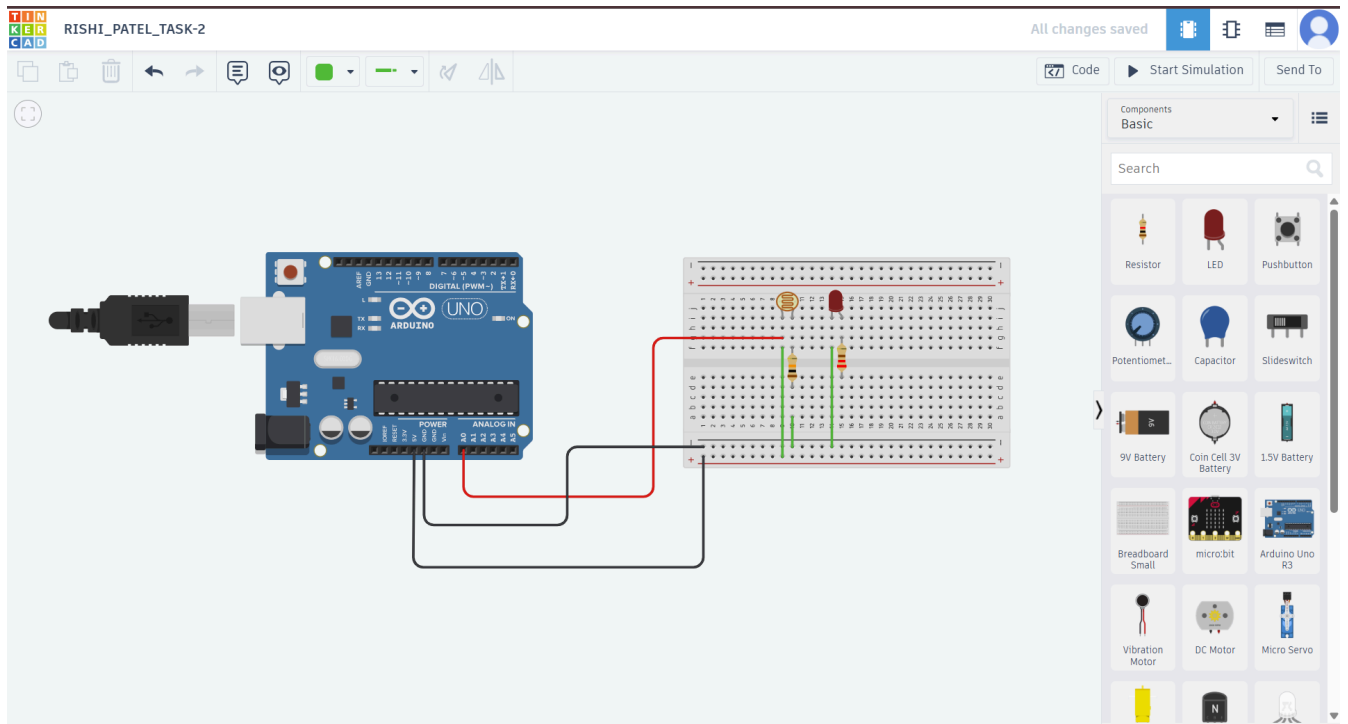
In This First We Have Taken An Arduino Uno R3 , Breadboard,Light,Two Resistors Of 10k Ohm And 220 Ohm And A Light Controller

Now We Will Make Connections :

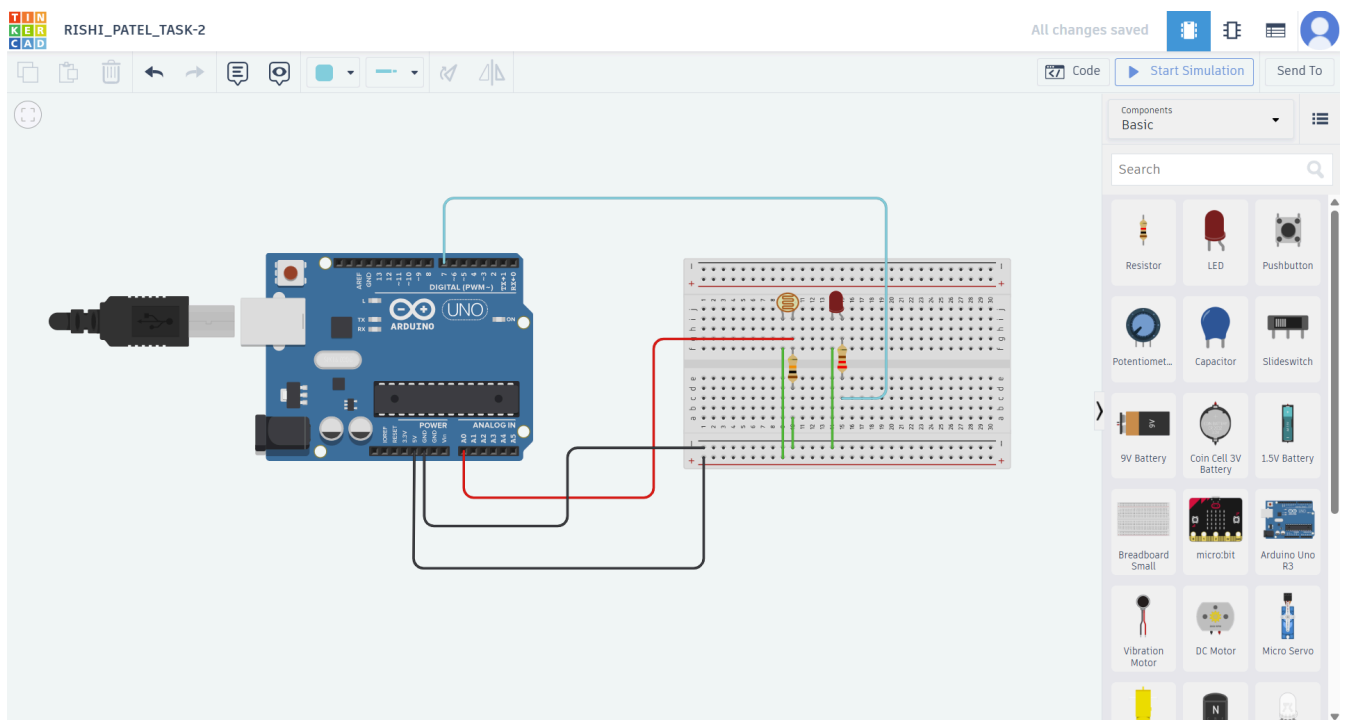








R



TIN KER CAD RISHI_PATEL_TASK-2 All changes saved

Code Start Simulation Send To

Text 1 (Arduino Uno R3)

```

1 const int ledPin = 7;
2 const int ldrPin = A0;
3
4 void setup() {
5   Serial.begin(9600);
6   pinMode(ledPin, OUTPUT);
7   pinMode(ldrPin, INPUT);
8 }
9
10 void loop() {
11   int ldrStatus = analogRead(ldrPin);
12   Serial.println(ldrStatus);
13
14   if (ldrStatus <= 80) {
15     digitalWrite(ledPin, HIGH);
16   }
17   else {
18     digitalWrite(ledPin, LOW);
19   }
20 }
21

```

Serial Monitor

TIN KER CAD RISHI_PATEL_TASK-2 All changes saved

Code Stop Simulation Send To

Simulator time: 00:00:16

Photoresistor
Name 1

Components Basic

Search

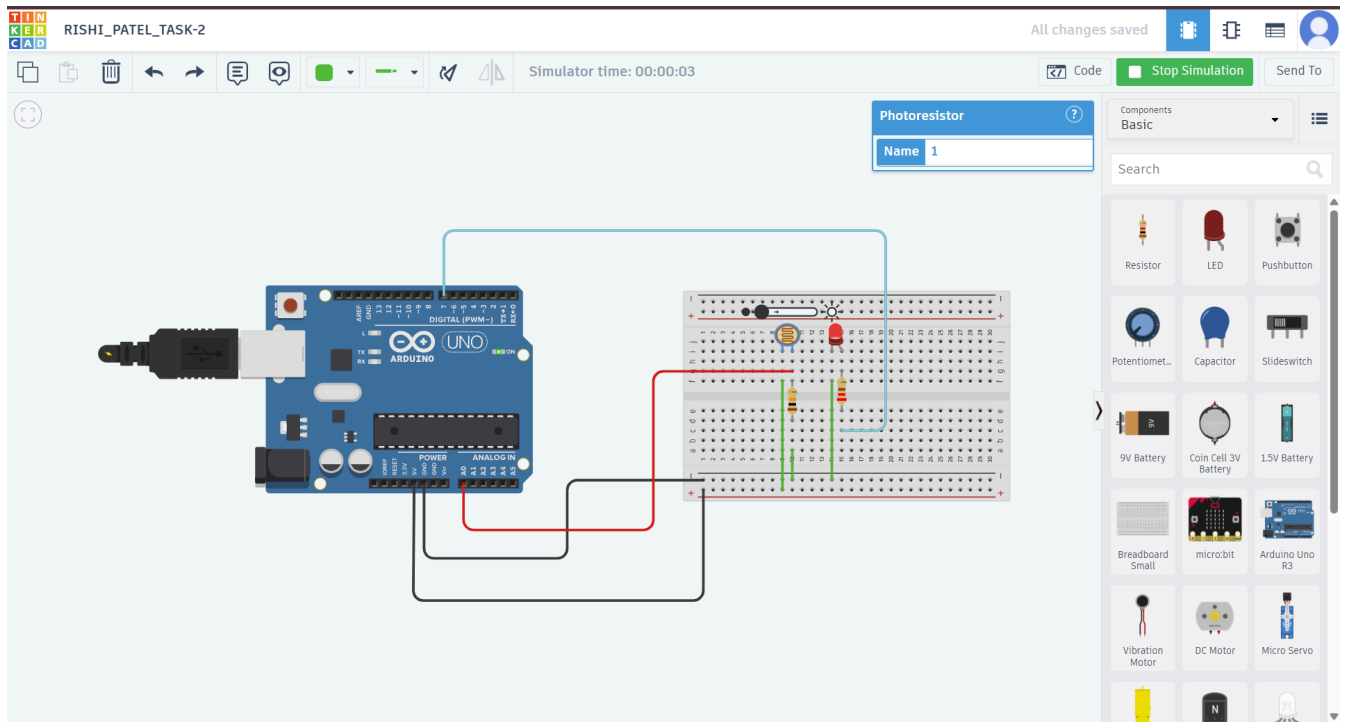
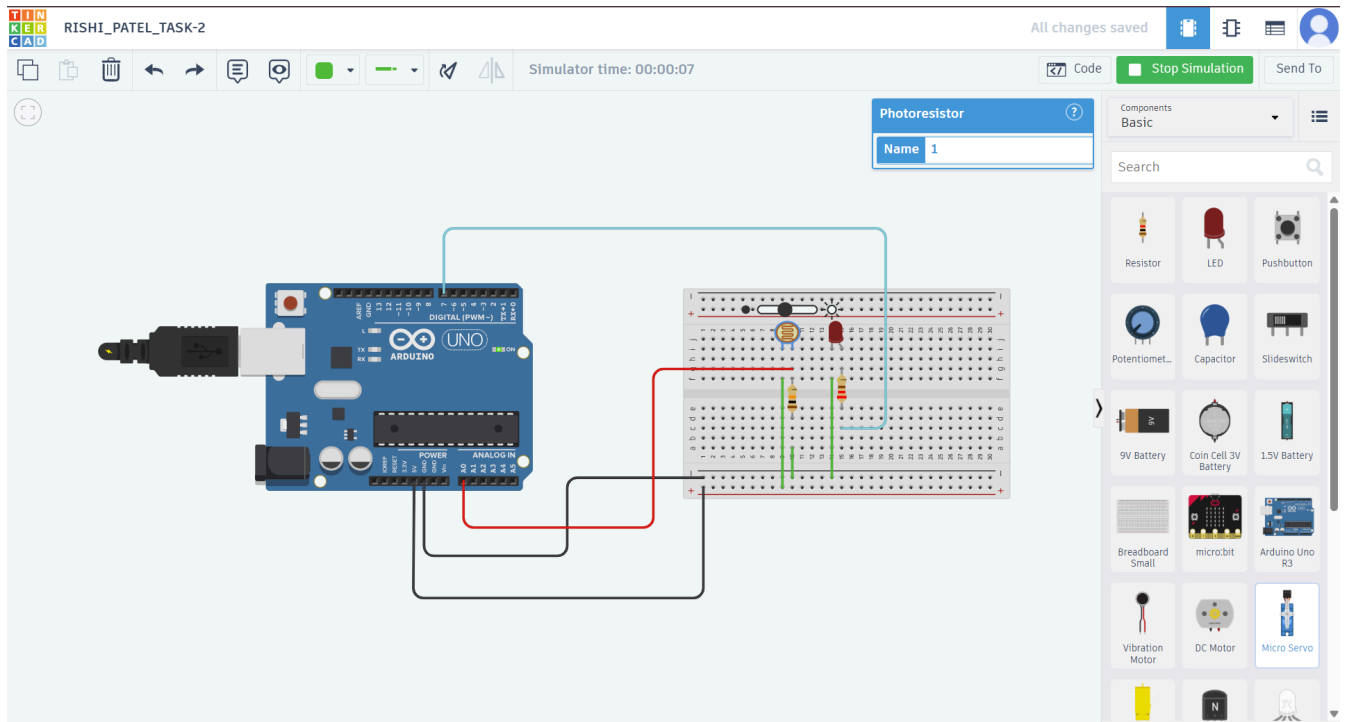
Resistor LED Pushbutton

Potentiometer Capacitor Slideswitch

9V Battery Coin Cell 3V Battery 1.5V Battery

Breadboard Small micro:bit Arduino Uno R3

Vibration Motor DC Motor Micro Servo



From The Above Running Circuit We Can See That When The Light When We Increase The Ldr The Light Gets Brighter And When We Decreases The LDR The Light Gets Deamer.

CODE:

```
const int ledPin = 7;
```

```
const int ldrPin = A0;
```

```

void setup() {
    Serial.begin(9600);

    pinMode(ledPin, OUTPUT);
    pinMode(ldrPin, INPUT);
}

void loop() {
    int ldrStatus = analogRead(ldrPin);

    Serial.println(ldrStatus);

    if (ldrStatus <= 80) {
        digitalWrite(ledPin, HIGH);
    }
    else {
        digitalWrite(ledPin, LOW);
    }
}

```

CODE EXPLANATION:

This Arduino program controls an LED using an LDR (Light Dependent Resistor).

- `ledPin = 7` defines the LED connected to pin 7.
- `ldrPin = A0` defines the LDR connected to analog pin A0.
- In `setup()`, serial communication is started and the pin modes are configured.
- In `loop()`, the Arduino continuously reads the light intensity from the LDR using `analogRead()`.
- The LDR value is displayed on the Serial Monitor using `Serial.println()`.
- If the LDR value is 80 or less (dark environment), the LED turns ON.

- If the value is greater than 80 (bright environment), the LED turns OFF.

Result: The LED automatically turns ON in darkness and OFF in bright light, simulating an automatic street light system.